

Ripple Map

An interactive explorer that uncovers the hidden knock-on effects of big events, wherever they show up, and shows exactly how sure we are about each one.

Executive and technical summary · Knock^On · Benjamin J. Sunter · bensunter.com/ripples · engine 6.1, with 6.2 in progress

WORKING DRAFT · 26 SEP 2026, 03:50 UTC · YELLOW BOXES ARE OPEN NUMBERS

PART I · EXECUTIVE SUMMARY

1 · THE VISION

"The entire point of the app is uncovering hidden impacts from upstream events, regardless of where that impact shows up. Attribution and traceability with statistical rigor are key."

"I'd rather this be a really cool interactive entertainment app than a p2p collaborative or competitive game." · "Balance rigor with product usability."

THE OWNER, 25-26 SEP 2026. EVERYTHING BELOW IS JUDGED AGAINST THESE THREE LINES.

In one sentence: Ripple Map takes an upstream shock (a hurricane, a heat wave, a model release, a rate decision), follows it across domains (jobless claims, grid demand, air travel, downloads, filings, prediction markets), and labels every step with a confidence tier that was earned by pre-registered tests, not asserted.

For example, a hurricane leads to disaster declarations, then jobless claims, then new business filings. Each step carries one of three words:

TIER	WHAT IT TAKES	HOW IT READS
Measured	Passes every test, including placebo and decoy controls and agreement from a second, independent forecast model	"Consistent with a ripple from ...; measured movement, not proof of cause"
Likely	Strong evidence, published with the measured rate at which findings like it turn out to be flukes	"About 1 in K findings at this level are flukes"
Watching	Pre-registered; its window is still open	"The mechanism exists; no measurable move yet"

The site never claims proof of cause. It is free, has no accounts, and is not an investment tool.

2 · WHAT MAKES IT DIFFERENT, AND HARD TO COPY

Tests are locked before the data is read. Every candidate link, its weight and its look schedule are frozen and written to an append-only, hash-chained ledger before any post-event number is used. The chain verifies after every change (696 rows, 54 freeze batches, 0 tests without a frozen candidate, as of 03:30 UTC).

Every link is tested against fakes. Three placebo families (fake dates, other events, look-alike series) plus daily decoy events run through identical code. Any tier a decoy earns is false by construction, so the fluke rate is measured, not assumed.

- A second opinion must agree.** A nightly BigQuery forecast check (`AI.CAUSAL_EFFECT`) must agree before anything is called Measured. On its first calibrated run it flagged 0 of 13 eligible decoys.
- Patterns across past events (engine 6.2).** Pooling past events of the same kind, and comparing hit regions with untouched ones, yields findings like "across N past hurricanes, jobless claims in hit states rose X %", each with a measured fluke rate.
- The failures stay on screen.** Stops that stayed flat, retractions, the decoy "control ripple" and failed launch gates are published. Nobody else publishes the denominator.

3 · RIGOR AND USABILITY, BALANCED

Strict tests favour obvious effects. The answer is not to loosen Measured but to add a plentiful, honestly labelled second kind of finding:

- Measured stays as strict as engine 6.1.** Engine 6.2 can only lift a statistical failure when a pre-registered regional contrast passes its own placebo test, and it adds a failure when a large family of past events disagrees.
- Patterns and Likely findings are the everyday product.** Each carries a strength word tied to a published fluke rate ("strong pattern", "pattern", "hint").
- Watching is part of the show.** Open windows, due dates and a reconstructed archive of past events give the map something to explore on day one.

4 · THE EXPERIENCE

An interactive entertainment explorer: the hero is the Ripple Map of today's shocks, each growing a cascade that crosses domains over days and weeks. Tapping a step opens an evidence card: mechanism, the series against its own seasonal normal, lag, fluke rate, agreeing sources and tier. Pattern cards summarise past events of the same kind; cascade cards are shareable; a light "call the next ripple" moment is optional.

Visual design: under redesign. The owner rejected the first frontend's look on 26 Sep. Three high-fidelity directions are being made on a real event (Hurricane Milton): **A** editorial seismograph, **B** radial ripple map, **C** chain of story cards. Chosen direction: `{{TBD: owner's pick of A / B / C}}`. The frontend is rebuilt on it afterwards.

- Superseded v5 content (kept out of this document).** Decision D-6 returned the project to the core vision, which removed:
- the daily puzzle as the main surface, slider scoring, points, streaks, streak shields, crowd stats and leaderboards;
 - the Split-Flap Riso art direction (petrol and marigold, departure-board flaps) as the final look; it is now one input to the redesign, not the answer.

5 · WHERE IT STANDS

35+ SOURCES COLLECTED DAILY	9,659 SERIES STORED	0 MEASURED HOPS SO FAR	377 / 500 MB DATABASE USED
AREA	STATE (26 SEP, 03:30 UTC)		
Data layer	Done. 35+ sources (54 registered) collected within written rules; YouTube added tonight. Wikipedia 3-year history: 360 of 996 pages filled, every 10 minutes.		

AREA	STATE (26 SEP, 03:30 UTC)
Engine 6.1	Deployed. All five audit blockers and 8 should-fixes fixed, plus a year-ago-anchored baseline for grid, travel, transit and prices. Acceptance runs in progress (1,931 pre-registered looks re-run so far).
Engine 6.2	Deployed; results pending. 29 family × outcome pairs and 4,035 past events frozen and hashed before any effect was computed. Patterns found: <code>{{TBD: n patterns / strong patterns of 29 pairs}}</code> .
Second opinion	Live. Nightly GitHub Action on BigQuery; wired into what the site publishes.
Publishing	Done. Daily frozen versions, feeds, share cards, data-contract tests pass.
Frontend	To rebuild on the chosen direction; code on branch <code>claude/admiring-goldberg-u36v1z</code> , not live yet.

6 • HONEST RISKS

1. **Rigor versus "hidden".** Strict tests favour obvious effects, and so far no hop of any kind has reached Measured. Engine 6.2 is the answer; whether it yields enough interesting patterns is `{{TBD: 6.2 result, n patterns with published fluke rate}}`.
2. **Power is low on some channels.** Before 6.1, a planted $+5\sigma$ effect was caught at most 29 % of the time. The 6.1 curve: `{{TBD: 6.1 spike-in recall at  $+3\sigma$  by channel}}`. The site never claims "no effect" beyond that curve.
3. **Live events start quiet.** Effects take days to weeks. The archive and pattern findings fill the gap.
4. **Outcomes are mostly US data.** Claims stay inside the measured geography.
5. **Free infrastructure.** 377 of 500 MB used, the scheduler stalls during long statements, and free projects pause after a week idle. Supabase Pro (about \$25/month) is recommended at launch.
6. **Design is not settled,** and one operator runs everything.

7 • WHAT'S NEXT

1. Finish the 6.1 acceptance runs and the 6.2 pooled results; publish both calibration sets.
2. Owner picks a design direction (A, B or C); rebuild the frontend on it, including pattern cards.
3. Wire the engine into the site end to end; three green live days.
4. Playtests with test users; final independent review.
5. Replace every yellow number in this document with the final build's value.
6. Owner merges to main to go live. Target date: `{{TBD: launch date}}`.

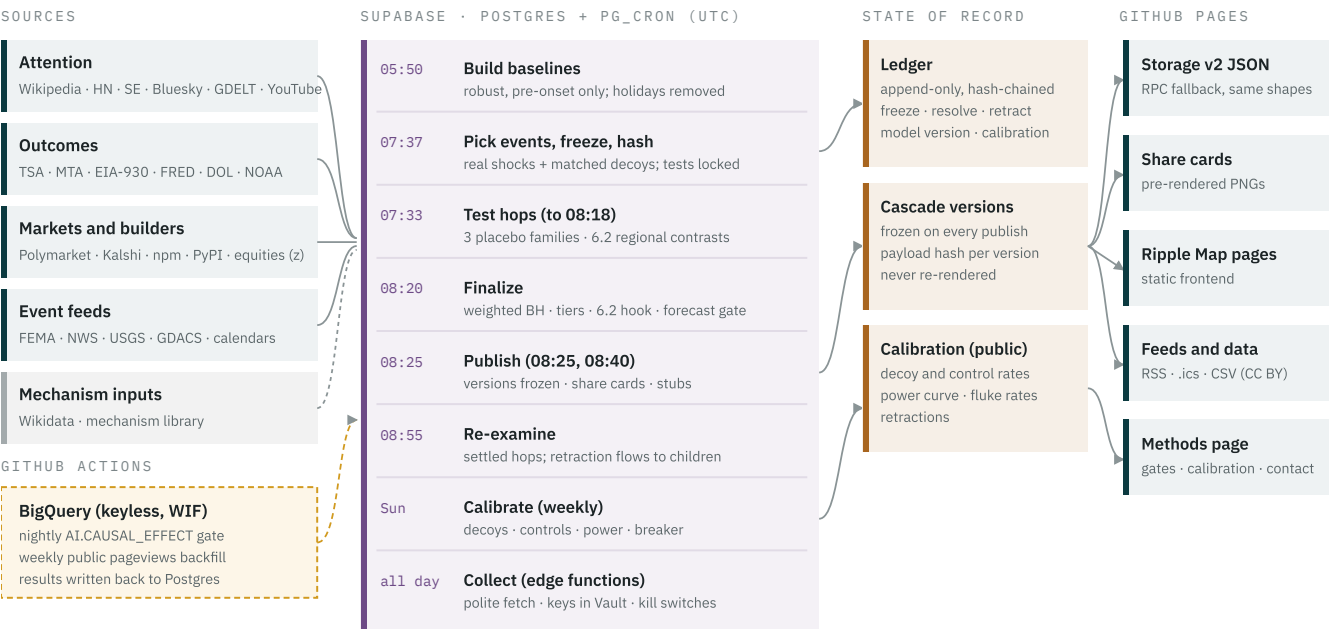
Owner asks: pick A / B / C; approve Supabase Pro at launch; merge. Non-blocking: the Stripe link for the pay-what-you-want support button (unlocks nothing; "Radar" alerts stay planned at \$49/month), legal review before any paid tier.

8 • MARKET GAP

Nearest products and where they stop: Ripple News (AI ripple explainers; narrative only, no baseline or null) · TickerTrends, Kalshi, Polymarket (first order, one domain; prices, not attribution) · RavenPack, Dataminr, Permutable AI (news to prices; no published denominators) · Futures Platform, Interos, Everstream (scenario wheels or one supply-chain domain; enterprise only). None combines live events, cascades past the second order, pre-registered placebo tests and public denominators. `{{TBD: re-verify competitors before sharing}}`

9 • ARCHITECTURE

Postgres is the engine; everything the public sees is a projection. Supabase edge functions collect daily aggregates into Postgres. The engine runs as SQL under `pg_cron` and writes an append-only, hash-chained ledger. GitHub Actions run the BigQuery work keylessly. The publisher freezes versioned JSON and share cards, and the site is a static GitHub Pages frontend. No visitor request ever reaches a source.



Grey inputs propose routes and never count as evidence. The dashed estimator is a second opinion: Measured requires it to agree.

Projections are disposable and can be regenerated. A published version is never edited; corrections become new versions or retractions. Every engine cron statement is kept short (under about a minute) because this instance's scheduler launches nothing while a long statement runs.

COMPONENT	RUNS ON	ROLE
Collectors	Supabase edge functions (Deno), triggered by <code>pg_cron</code>	One polite-fetch path for every outbound request; aggregates only
Engine	Postgres SQL and PL/pgSQL under <code>pg_cron</code> , chunked	Baselines, freeze, tests, placebos, tiers, retraction, 6.2 pooling
Secrets	Supabase Vault; GitHub secrets and variables	Read server-side through a <code>service_role</code> -only accessor
Second opinion, history	GitHub Actions → BigQuery via Workload Identity Federation	Nightly forecast gate; weekly Wikipedia pageview backfill. No service-account keys exist
Frontend	GitHub Pages (static), Supabase Storage JSON	Reads frozen payloads only; no accounts

10 • DATA LAYER

Every source has a row in the source registry with a demarcation tier, a kill switch and the surfaces it may feed. Turning a source off removes its data from every page on the next build.

SOURCE	GIVES	KNOWN WEAKNESS	TIER
Wikipedia pageviews	Daily readers per article; 3-year history (360 of 996 filled)	Attention, not behaviour; REST is rate-limited, so bulk history comes from BigQuery's public copy	G · CC0
YouTube Data API (<i>new</i>)	Most-popular charts, 40 countries × 7 categories: view sums, new entrants, velocity	Chart-level only; 280 of 10,000 daily quota units	G · key
HN, Stack Exchange, Bluesky, Mastodon, GDELT, TV	Chatter and coverage by term	Short history (warm-up); a channel that proposed a link cannot confirm it	G / Y
EIA-930, TSA, MTA, Citi Bike	Hourly grid demand by region (2015 on); daily air travellers; NYC transit	Air travel is national only; transit is New York only	G
FRED, BLS, DOL, Census BFS	Rates, prices, jobs, weekly state jobless claims, business applications	Monthly or weekly lag; revisions (vintages stored)	O key / G
NOAA, FEMA, NWS, USGS, GDACS	Weather baselines; declarations, warnings, quakes: shocks and institutional responses	US-centric; declarations lag events	G
npm, PyPI, GitHub, Hugging Face	Builder adoption: downloads, stars	CI-bot inflation; npm history starts mid-2022	G / Y
Polymarket, Kalshi; Twelve Data, Alpha Vantage	Prediction markets; sector moves as derived z-scores	Thin markets; raw prices never republished, no tickers in open data	Y / O*
SEC EDGAR full-text, USAspending	Filings and federal spending that mention an event	Capped at 200 EDGAR requests/day	G
Wikidata, mechanism library	Typed relations that propose routes	Proposes only; never counts as evidence; read as of the event date	G · CC0

Demarcation: the lines we don't cross

Creative means more independent, legitimate signals, not grey access. Each source is graded by a written test (official channel? barrier? robots.txt? terms? personal data? published rate limit?), and the answer is recorded.

TIER	MEANING	USE
GREEN	Official or openly licensed; no barrier; robots.txt allows; terms permit	Free and paid, within the licence
YELLOW	Allowed but undocumented, non-commercial or third-party	Derived only, free product, kill switch
ORANGE / *	Issued key or permission; * = a documented keyed API on a host whose robots.txt disallows it, by owner decision	Per agreement
RED	A barrier, a robots.txt deny with no key route, or terms that ban it	No request at all

- **Honest identity.** A declared User-Agent with a contact URL. The Wikimedia contact page is live at bensunter.com/ripples/methods/; every Wikimedia collector stops if it cannot be reached.
- **Stop, don't retry.** A 429 or 503 stops the host until the next UTC midnight; a 401 or 403 stops it until the owner clears it. No CAPTCHA solving, proxies, IP or key rotation, or logged-in scraping.

- **Aggregates only.** Counts, ranks and our own z-scores; no post text, handles or article bodies.
- **YouTube retention.** No per-video row is ever written. The only raw cache is today's and yesterday's chart snapshot (always under 48 hours old); a daily job deletes any raw row older than 28 days as a safety net, inside YouTube's 30-day cached-data rule. Titles are read in memory only.
- **Documented exceptions.** Wikimedia's API and SEC EDGAR full-text search (2 req/s, 200/day) skip the robots.txt check; every other rule applies.

Compliance audit, 25 Sep: pass after three fixes in the older pipeline. Re-audit on the final build: `{{TBD: final per-host compliance re-audit}}`. The demarcation map is an engineering risk map, not legal advice.

11 • THE CASCADE ENGINE

A typed mechanism graph decides what to test. An event study with placebo controls decides whether the downstream series moved. Pooling across past events decides how much to trust a class of link. Engine versions: 6.0 (built), 6.1 (independent audit fixed), 6.2 (patterns and regional contrasts, in progress).

6.0: the core

- **Shocks and decoys.** The day's real shocks each get a family (hazard, tech, policy, media, person). Matched decoy events, calm over the previous 14 days, go through identical generation, tests and chaining.
- **Candidates, frozen first.** Family mappers (storm → state → claims, grid demand, travel, declarations), the mechanism library and Wikidata propose at most 60 paths per event. Paths, weights and look dates are hashed into the ledger before any post-onset number is read.
- **Baselines.** Each series is compared only with itself: logged counts, shrunk weekday effects, a year-ago term, common-shock demeaning across a source's panel.

$$AR_t = (\tilde{x}_t - \hat{y}_t) / \sigma - c_t \quad \cdot \quad \sigma = 1.4826 \cdot MAD \quad \cdot \quad c_t = \text{the day's median } z \text{ across the panel}$$

- **Three placebo families.** Date (fake onsets of the same pair, season-matched), link (other events paired with the same series), topic (look-alike series at the real onset). A hop's p is the maximum across families; fewer than two families caps it at Likely.
- **Multiple testing.** One weighted Benjamini–Hochberg run per day covers real and decoy hops together, with weights frozen before data.
- **Tiers.** Measured needs $q \leq 0.05$ at the final pre-registered look, an outcome channel, two independent channels, correct lag order, no common-shock day, no pre-trend, and the forecast gate. Attention-only hops are capped at Likely. Likely shows its one failing condition by name.
- **Chaining.** Only Likely-or-better hops get children; a fork test draws a direct path as a sibling; nothing is multiplied along a chain for display.

6.1: the independent audit and its fixes

An independent statistician found the engine *safe by absence of power* (nothing had ever reached Measured) but with four anti-conservative holes that would matter once power was restored, plus a deploy gate that could not pass. All five blockers are fixed and deployed; the ledger still verifies bit-identically.

#	AUDIT FINDING	FIX IN 6.1
B1	Positive-control fixtures could not reach their targets	Controls look up geo-annotated events; regional expectations (affected grid regions, state declarations); attention targets no longer expected to be Measured
B2	The null scale was computed over post-onset and regime periods, and was degenerate on zero-heavy series	Robust "normal" from windows that end before the event (minus a buffer), with floors; degenerate series excluded
B3	Seasonal nulls unmatched outside physical and economic series	Season-matched fake dates on every channel; series under two years old cannot be Measured
B4	Holiday and common-shock flags never fired	US holidays, Christmas week and Black Friday registered (759 days); panel-wide days flagged; state filters honoured
B5	Settled hops were never re-examined; retraction did not propagate	Daily re-examination at 08:55 in the hop's own frozen family; retraction and demotion flow to children

Also fixed: duplicate-path removal, the mechanism graph read as it stood at the event date, honest fluke-rate labels while warming up, price series as levels, and (after the report) a year-ago-anchored baseline for grid, travel, transit and price series. Not yet done: first-difference baselines for growth series, a link-family redefinition, four minor items.

6.2: patterns across events, and hit versus untouched regions

The aim is many interesting, honestly labelled findings without loosening Measured.

- **Family pooling.** Each frozen (event family × outcome) pair pools every past event of that kind. Per-event effects are standardised against 40 season-matched fake onsets, combined by random-effects meta-analysis, and judged by a placebo pool (300 replicates, each pooling one fake onset per real event), not by a normal approximation. Dose–response against event magnitude is reported as exploratory. The batch is one Benjamini–Hochberg family.
- **Regional contrasts.** Difference-in-differences of hit regions against every untouched region (state claims, business applications, grid regions, declarations, weather), with a three-lead pre-trend test and 30 random in-space placebo groups; synthetic control with a rank test against 20 placebo groups where flagged. Holidays, national news and nationwide weather cancel; a shock that also hits the comparison regions does not.
- **New events borrow strength honestly.** A live event's hop shows an empirical-Bayes shrunk estimate next to its own evidence and the family's, with a fixed note that the family result is not evidence about this event.
- **Tiering hook.** A passing regional contrast can lift only statistical failures (q, placebo family, single channel, final look not reached); it never lifts a common-shock day, pre-trend, rival, reversal, parent or young-series failure. A family of 8+ past events that disagrees adds a Measured failure.

STRENGTH WORD	RULE (PRE-REGISTERED)	PUBLISHED FLUKE RATE
strong pattern	$q \leq 0.05$ and the pre-registered sign	<code>{{TBD: measured fluke rate, strong}}</code>
pattern	$q \leq 0.20$ and the pre-registered sign	<code>{{TBD: measured fluke rate, pattern}}</code>
hint	placebo $p \leq 0.05$ but not surviving BH	shown as exploratory
too few events	fewer than 8 past events	not published as a pattern

Pre-registration: ledger seq 931 (method, config, tier rule) and seq 932 (29 pairs, 4,035 events, every panel's extent), hashed before any effect was computed. Example result for the headline pair: `{{TBD: storms → state jobless claims, +X % over 4 weeks, N events, fluke rate}}`.

12 · SECOND OPINION: THE FORECAST GATE

What it is. BigQuery's preview `AI.CAUSAL_EFFECT` fits a univariate forecast for one intervention date and returns the effect with an interval. **What it is not.** No discovery, no common-shock control, no multiple-testing control, so it cannot replace the engine. It is used only as an independent check.

- **Measured requires agreement:** matching sign and an interval that excludes zero at the pre-registered horizon. The gate is wired into what the site publishes: a demoted hop shows the gate's reason ("forecast check pending", "disagrees", "cannot assess this series").
- **Eligibility rule after the first run.** The first run called 51 of 81 decoys "significant", almost all from zero-heavy declaration counts (the forecast interval collapses to zero) and 91-day horizons fitted on too little history. A series now counts only if its interval has positive width and the horizon is at most 29 days; a hop with no eligible series cannot be Measured.
- **Wording:** "vs forecast (range)", never "probability of causal effect". **Cost guard:** 204.8 GiB/day quota cap, \$10 budget alert, partition pruning, dry-run bytes logged.

FIRST CALIBRATED RUN (ELIGIBLE HOPS)	FLAGGED	READING
Decoys	0 of 13	No false alarms after the eligibility rule (51 of 81 before it)
Negative controls	4 of 54 (0 agree)	Significant but never in the expected direction, so none would pass the gate
Positive controls	3 of 20 agree	A conservative gate: it rarely confirms, which costs Measured hops, not honesty

13 · CALIBRATION AND ACCEPTANCE

Each number below is published on the methods page. "Before" values are from the audit of engine 6.0; "now" values are from the running 6.1 recompute and will change.

CHECK	TARGET	RESULT
Ledger integrity	Chain verifies; every frozen batch recomputes	Pass · 696 rows, 54 batches, 0 tests without a candidate
Null scale (TSA, MTA)	Spread within 0.7–1.7	Pass · 1.13 / 1.16 and 1.14 / 1.59
Holiday trap (Thanksgiving storm)	Never Measured	Pass by construction; end-to-end run <code>{{TBD: Thanksgiving e2e result}}</code>
Decoys, Likely or better	$\leq 10\%$, upper bound $\leq 15\%$; Measured 0 on ≥ 500 final looks	Before: 85 of 505 (16.8 %). Interim 6.1: 0 of 46. Final: <code>{{TBD: decoy Likely+ rate, n, Wilson bound}}</code>
Negative controls	Within the decoy band	Before: 7 of 90. Final: <code>{{TBD: negative-control rate vs decoy band}}</code>
Positive controls (6)	Realistic ones reach Measured	Before: 0 of 6. Interim 6.1: 0 of 6 (recompute unfinished). Final: <code>{{TBD: tier per positive control}}</code>
Power (planted effects)	$\geq 50\%$ recall at $+3\sigma$, lag ≤ 3	Before, at $+5\sigma$: physical 29 %, jobs 16 %, builders 0 %. 6.1: <code>{{TBD: 6.1 spike-in curve}}</code>
Null uniformity	KS $p \geq 0.05$ on 500 held-out pairs	<code>{{TBD: KS p}}</code>

CHECK	TARGET	RESULT
6.2 decoy families	Placebo-pool fluke rate published per strength word	{{TBD: 6.2 decoy/placebo calibration row}}

The honest expectation from the audit: realistic Measured passes are regional (hurricane → affected grid regions and state declarations) and builder downloads after open-model releases; ChatGPT → npm is untestable (too little pre-launch history), and the Texas heat dome depends on the new year-ago baseline.

14 · PUBLISHING, CONTRACTS AND SECURITY

- **Publishing.** Daily frozen cascade versions with payload hashes, mirrored JSON in Storage, pre-rendered share cards, RSS and `.ics` feeds, open CSV (CC BY, GREEN-derived fields only). Shared links point to frozen versions.
- **Contract tests.** `rm_contract_test()` checks every public payload shape and the Measured wording; the forecast gate has its own tests. Engine unit tests T1–T25 cover freeze identity, placebo code paths, holidays, frozen look dates and the 6.2 hook (which may lift only the allowed failures). Latest full pass: `{{TBD: engine test harness T1-T25 pass record}}`.
- **Database.** Row-level security on every table. Public RPCs are read-only, `SECURITY DEFINER`, empty search path, IDs-only inputs. Final advisor run: `{{TBD: Supabase advisor ERROR / WARN counts}}`.
- **Secrets.** API keys live in Vault and GitHub secrets, never in the repo, logs or payloads; scoped credentials go only to their own hosts. Google Cloud uses Workload Identity Federation restricted to this repository, with BigQuery job-user and data-viewer roles only.
- **Visitors.** No accounts, no tracking pixels; a hashed client ID with a daily salt.
- **Integrity.** Tamper-evident, not tamper-proof: the chain head is not yet anchored outside the database.

15 · HONESTY GUARDRAILS

We can claim

- Timing: the series moved after the shock, with the lag
- Size against the series' own normal, shrunk, with an interval
- Rarity against calibrated placebos and decoys
- How many independent channels agree
- Patterns across N past events, with a measured fluke rate
- Hit regions versus untouched regions

We never claim

- "Caused", "drove", "because of"
- A share of an effect, dollar or job amounts
- Any probability for a whole chain
- "No effect" beyond the published power curve
- That a family pattern proves this event's effect
- Tickers, trades or price forecasts

Numbers come only from SQL. Reconstructed archive lines are labelled on every surface and excluded from live denominators. Hazards with casualties get neutral copy and no teaser card.

16 · INFRASTRUCTURE CONSTRAINTS

CONSTRAINT	NOW	MITIGATION
Supabase free tier, 500 MB	377 MB at 03:30 UTC (self-imposed cap 400 MB); collectors refuse writes near the cap	Compact array store for history (3 years of Wikipedia ≈ 4.4 MB); retention jobs; Pro recommended at launch

CONSTRAINT	NOW	MITIGATION
Scheduler stalls	<code>pg_cron</code> launches nothing while any single long statement runs; collectors pause too	Every engine job chunked to under about a minute; long harness and power runs scheduled in quiet windows
Auto-pause	Free projects pause after a week idle	Daily jobs keep it active; Pro removes the risk
BigQuery cost	Nightly and weekly Actions only	204.8 GiB/day cap, \$10 alert. Typical nightly scan: <code>{{TBD: GiB per nightly run}}</code>

17 • LAUNCH GATES

Positive controls block an engine deploy. Every other failing gate is printed on the methods page with its automatic fallback.

GATE	CRITERION	RESULT
G1 Positive controls	Realistic controls Measured on the archive	<code>{{TBD: G1 positive-control result}}</code>
G2–G4 Decoys, negatives, null	As in §13	<code>{{TBD: G2–G4 decoy, negative, null results}}</code>
G5 Forecast gate calibrated	Decoy flag rate published; Measured requires agreement	Pass • 0 of 13 decoys
G6 Freeze integrity	Ledger verifies; no test without a frozen candidate	Pass • 03:30 UTC
G7 Pattern calibration	6.2 fluke rates published per strength word	<code>{{TBD: G7 pattern fluke rates published}}</code>
G8 Copy audit	No chain probability, ticker, dollar figure or causal verb anywhere	<code>{{TBD: G8 copy audit result}}</code>
G9 Budgets	DB under 400 MB; daily run done by 08:18	<code>{{TBD: G9 MB, finish time}}</code>
G10 Three live days	End to end, collect → publish → share cards	<code>{{TBD: G10 n of 3}}</code>
G11 Frontend	Chosen direction built; shell ≤ 120 KB; fast on a 360 px phone	<code>{{TBD: G11 frontend built and budget}}</code>
G12 Decommission	v5 game RPCs revoked; no points, streaks or crowd stats anywhere	<code>{{TBD: G12 decommission check}}</code>

18 • ROADMAP

WHEN	WHAT
Before launch	6.1 acceptance, 6.2 results, design pick and rebuild, integration, playtests, final review
Launch <code>{{TBD: launch date}}</code>	Ripple Map with the reconstructed archive and pattern cards as the day-one hero; owner posts by hand for 14 days
Days 28–90 of collection	Warm-up sources (news, TV, Bluesky, charts, YouTube, Kalshi) become eligible to count
After launch	Growth-series baselines; link-family redefinition; anchoring the ledger head outside the database; clickstream and Trends when available
Before any paid tier	Move YELLOW non-commercial sources off paid surfaces; legal review

A · Symbols

SYMBOL	MEANING
AR, σ	Abnormal response against the series' own baseline, and its robust scale
P_h, q_w	Placebo p (max over families); weighted Benjamini–Hochberg value over the day's hops, real and decoy together
f	Decoy-calibrated fluke rate ("flukes about 1 in K")
$d_{pool}, \tau^2, p_{placebo}$	6.2 pooled family effect, between-event variance, and its placebo-pool p
P_{space}, P_{pre}	6.2 in-space placebo rank of a regional contrast; pre-trend check (flat if ≥ 0.10)

B · Sources for this document

STATUS (26 Sep, 03:15 UTC), OWNER_DECISIONS D-1 to D-13, ENGINE_SPEC (6.0), ENGINE_AUDIT, ENGINE_FIXES (6.1), ENGINE_62 (design; results pending), EXPERIENCE_SPEC (visual design superseded), ATTENTION_STACK, DEMARCATION, attention COMPLIANCE (incl. YouTube), BigQuery eligibility and publish-gate SQL, WIKI_HIST, and live database reads at 03:30 UTC (size, series count, ledger verification, forecast-gate calibration view, recompute state).

C · Open numbers

1. §4	owner's pick of A / B / C	15. §13	6.1 spike-in curve
2. §5	n patterns / strong patterns of 29 pairs	16. §13	KS p
3. §6	6.2 result, n patterns with published fluke rate	17. §13	6.2 decoy/placebo calibration row
4. §6	6.1 spike-in recall at $+3\sigma$ by channel	18. §14	engine test harness T1–T25 pass record
5. §7	launch date	19. §14	Supabase advisor ERROR / WARN counts
6. §8	re-verify competitors before sharing	20. §16	GiB per nightly run
7. §10	final per-host compliance re-audit	21. §17	G1 positive-control result
8. §11	measured fluke rate, strong	22. §17	G2–G4 decoy, negative, null results
9. §11	measured fluke rate, pattern	23. §17	G7 pattern fluke rates published
10. §11	storms → state jobless claims, +X % over 4 weeks, N events, fluke rate	24. §17	G8 copy audit result
11. §13	Thanksgiving e2e result	25. §17	G9 MB, finish time
12. §13	decoy Likely+ rate, n, Wilson bound	26. §17	G10 n of 3
13. §13	negative-control rate vs decoy band	27. §17	G11 frontend built and budget
14. §13	tier per positive control	28. §17	G12 decommission check
		29. §18	launch date